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Development of a model to predict die filling under suction on different rotary tablet presses

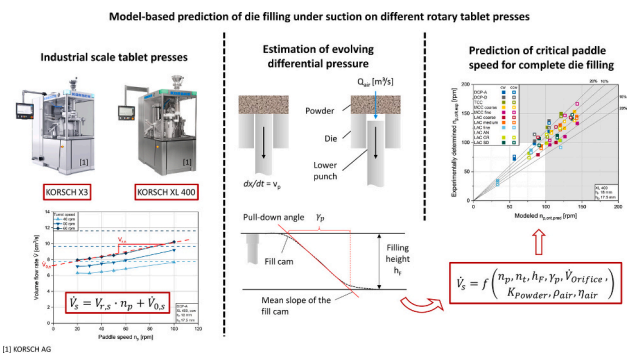
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HIGHLIGHTS

- Prediction of die filling based on powder properties and process parameters.
- Correlation between the die filling performance and the estimated differential pressure.
- Prediction of paddle speed for complete die filling with an accuracy of approx. 20 %.

GRAPHICAL ABSTRACT



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ABSTRACT

Die filling is a key step in tableting, as it directly affects tablet weight. However, the relationship between process parameters, material properties and filling performance remains poorly characterized, often requiring extensive experimentation and material use during formulation process development as well as scale-up. This study investigates suction filling of eleven different commonly used excipients using two production-scale rotary presses with single paddle feed frames. As a basis, differential pressure inside the dies, generated during the punch pull-down, is calculated based on rational assumptions. Following, a correlation between this differential pressure and the volume flow rate into the dies was identified and used to develop a process model. This model builds up on a previously developed approach for gravity filling, aiming to predict the minimal paddle speed for complete die filling under suction based solely on material properties and process parameters. Powder permeability emerges as a key factor, with lower permeability amplifying the suction effect. The model predicts critical paddle speeds with medium accuracy, generally within 20 % error providing a robust basis for future process design and optimization. Importantly, it demonstrates that a complex filling mechanism can be reliably predicted using only simple and widely available powder characterization methods without the need of experiments on rotary presses. This makes the approach particularly valuable for industrial application, as it reduces the reliance on extensive trial-and-error experiments and accelerates the efficient development of feasible formulations and robust manufacturing processes.

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1. Introduction

For the industrial production of pharmaceutical tablets, rotary tablet presses are used. Here, die filling is achieved by using a feed frame, typically equipped with rotating paddle wheels, which force the powder to flow into the dies. Following, the powder is compacted to a tablet and ejected from the die. As die filling predominantly defines the final tablet weight and weight variability, representing critical product properties, it is a crucial process step. Until today, several challenges during production are caused by a non-optimal filling process. Therefore, it is important to understand influencing factors in depth to ensure a stable and reliable process to meet the required quality and to enable process optimizations.

One strong influencing factor on the filled powder weight is the used filling mechanism. Two filling mechanisms can be distinguished – gravity and suction filling. During gravity filling, the lower punch enters the feed frame while being in the lowest position, so particles fall due to gravitational forces into a die filled with air. The entering powder flow is hindered by escaping air, which can cause turbulences and lead to particle segregation [1,2]. In order to investigate and observe the filling process in detail, Wu et al. developed a linear feed shoe system. Several studies on linear feed shoe systems focused on the understanding of flow patterns, effects of process parameters, and the impact of material properties [1,3–8]. Additionally, studies were conducted on rotary systems and rotary tablet presses to incorporate prior neglected influencing factors, like the turret rotation or effects based on the rotating paddle wheels inside the feed frame [8–13]. Recently, a model was developed by the authors to predict the necessary paddle speed to reach a desired tablet weight for gravity filling at a rotary tablet press in pilot-scale with a single-paddle feed frame [12]. Additionally, the model was extended enabling the scale-up to larger rotary presses [13]. The filling time t_{fill}, which depends on the open filling angle of the feed frame and the turret speed, showed a strong impact on the filled powder weight. Consequently, it was incorporated into the model by dividing the mean tablet weight $\overline{m}_T$ by t_{fill} and the bulk density ρ_b to derive the powder volume flow rate into the dies. This volume flow rate linearly depends on the paddle speed (expressed as its rotation frequency 1/min) n_p, depicted by the linear die filling model for gravity filling. The linear model consists of the slope, also referred to as the rotation-specific fill volume V_r and the axis intercept, referred to as the reference volume flow $\dot{V}_0$:

$$\dot{V} = \frac{\overline{m}_T}{\rho_b \cdot t_{fill}} = V_r \cdot n_p + \dot{V}_0 \tag{1}$$

For the investigated materials, a high bulk density and a high permeability were favourable for die filling. Moreover, the impact of the paddle speed on the tablet weight was stronger for materials with a low Hausner ratio and a low dynamic angle of repose, which represent low consolidation and good flow properties.

During suction filling, in contrast, the lower punch is pulled downwards below the powder bed inside the feed frame, which results in a differential pressure [2]. The positive impact of suction filling due to the negative differential pressure has been shown in numerous studies in the literature [2,5–7,14–17]. Some studies were carried out on model linear feed shoe systems at which the lower punch was pulled down at different speeds [2,5–7,14]. Jackson et al. first highlighted the considerable positive impact of suction filling on the filled powder weight [2]. The authors let the lower punch drop at that moment it was fully covered with powder in the feed frame: For microcrystalline cellulose, the maximum feed shoe velocity, which enables complete die filling of a model linear feed shoe system (critical shoe velocity), was increased by a factor of 2.5. Moreover, the decline of the fill ratio above the critical shoe velocity was found to be less steep for suction filling. Due to the powder above the die, air cannot enter the die fast enough during the volume expansion, so a negative differential pressure against ambient conditions develops. Air in the powder's pores and from above expands into the die, accelerating the particles, which results in a considerably higher fill ratio compared to gravity filling [2]. Consequently, suction filling is commonly used on rotary presses during the production of pharmaceutical tablets. Only for complex tablets such as multi-layer or press-coated tablets, gravity filling is applied for the second layer on the tamped first layer(s), and thus the lower punch is in a fixed position.

Jackson et al. observed a column-like flow into the die during suction filling, which reduces segregation [2]. Later on, this plug-flow like flow was confirmed by other authors for different powders, e.g. with high-speed cameras or in coupled CFD/DEM simulations [14,15]. Sinka et al. improved the system by installing an electrical drive for the lower punch to investigate specific pull-down velocities [5].

The effect of suction filling on the fill ratio depends on powder properties [6]. Fine cellulose grades present a strong improvement of their filling behavior, while a coarser grade fills only slightly better than during gravity filling [6]. Mills et al. proposed the permeability of the powder bed to be the responsible powder property [6]. The lower the permeability, the lower the airflow through the powder during the pull-down of the lower punch. Therefore, the evolving differential pressure is higher. An improvement in filling degree for a formulation with an increasing ratio of fine cellulose was also observed at a rotary press, using suction filling [18]. The authors explained their finding by an increased amount of air in the formulation, resulting in an improved dynamic flowability. The dependency of the mass flow rate under differential pressure on the particle size was also confirmed on an experimental test rig with applied differential pressures [17]. Based on these observations, Baserinia et al. developed a model to predict the mass flow rate based on the applied differential pressure for powder flow through an orifice from a bin [17].

Additionally, the pull-down velocity presents a strong impact on the resulting filling level. A faster punch velocity can be related to higher fill ratios for fine and coarse powders [6,14]. However, it is important to guarantee a sufficient powder supply from the feed frame. For a very fine cellulose grade, a decline in fill ratio was identified for very high pull-down velocities, which was explained by the very low flowability [6]. Zakhvatayeva et al. found not only a dependency on the punch velocity, but also for the combination with the shoe velocity. For a cohesive material, the fill ratio was rising with increasing ratio of shoe to punch velocity until the shoe velocity was so high that the punch still moved downwards although the feed shoe already passed. A densification of the powder with increasing shoe velocity due to inertia, leading to a lower effective permeability, was used as explanation. Conversely, for coarse, free flowing powders the highest fill ratio was found for the lowest shoe to punch velocity ratio, decreasing linearly with a rising ratio. Here, a higher particle inertia, reducing vertical particle velocity, can be responsible [14].

At high punch velocities and moderate shoe velocities in a linear feed shoe system, the lower punch can reach the lowest position while the feed frame is still above. For rotary presses, this is always the case, as the dosing unit follows the fill cam at a fixed position, which is necessarily still below the feed frame. Consequently, the air pressure in the die can rise as powder still enters but the volume does not expand anymore, leading to a pressure increase as during gravity filling [6]. In order to investigate this, Baserinia et al. installed a differential pressure sensor in the face of the lower punch to monitor the pressure de- and increase during a linear filling process [7]. For a fine cellulose, an increase in air pressure was shown for a low shoe speed of 100 mm/s, while it did not appear at the highest investigated speed of 300 mm/s. Based on their findings, Baserinia et al. extended a model for gravity filling, developed by Schneider et al., to predict the delivered powder weight during gravity and suction filling with respect to the evolving pressure gradients on a linear feed shoe model system [4,7]. However, for each material, several constants have to be determined beforehand based on experiments with that specific system. Although the authors apply the model, calibrated at a model linear feed shoe system, to a rotary press, it remains unclear if the assumed values of the air pressure during gravity

and suction filling, as well as the values for the constants stay the same for the rotary press. At the present time, measurements of differential pressures inside the dies have not been established on rotary tablet presses, which is why it is not possible to make a precise distinction between gravity and suction filling during the die filling process. It is therefore unclear to what extent suction filling is superimposed by gravity filling.

The studies mentioned above were conducted on simplified test rigs and are therefore only applicable to a limited extent to real rotary presses used in the production of pharmaceutical tablets. Until now, there has been a lack of in-depth understanding of how the process parameters on a rotary press affect the tablet weight. An increase in turret speeds leads to a shorter filling time on one hand, which limits the powder flow. On the other hand, a faster turret speed results in a higher pull-down velocity of the lower punch, as it passes the fill cam more quickly. This can lead to a higher differential pressure and thus an enhancement of the powder flow. This study aims to improve the basic understanding of the suction filling mechanism on rotary tablet presses. Therefore, basic physical considerations are made to estimate the differential pressure inside the die. Following, a model is presented which enables the prediction of a minimal, critical paddle speed to fill the dies completely at a certain turret speed based on powder properties.

2. Materials and methods

2.1. Materials

In this study, eleven pharmaceutical excipients were used: three sieved lactose monohydrate grades (LAC coarse, Pharmatose® 90M; LAC medium, Pharmatose® 100M; LAC fine, Pharmatose® 125M, all from DFE Pharma, Goch, Germany), spray-dried lactose monohydrate (LAC SD, SuperTab® 11SD, DFE Pharma, Goch, Germany), agglomerated lactose monohydrate (LAC GR, SuperTab® 30GR, DFE Pharma, Goch, Germany), anhydrous lactose (LAC AN, SuperTab® 21AN, DFE Pharma, Goch, Germany), tricalcium citrate (TCC, Tricalcium Citrate, Jungbunzlauer, Ladenburg, Germany), anhydrous dicalcium phosphate (DCP-A, DI-CAFOS® A150, Chemische Fabrik Budenheim KG, Budenheim, Germany), dicalcium phosphate dihydrate (DCP-D, DI-CAFOS® D160, Chemische Fabrik Budenheim KG, Budenheim, Germany), microcrystalline cellulose in a fine grade (MCC fine, Emcocel® 90M, JRS Pharma Rosenberg, Germany) and in a coarse grade (MCC coarse, Vivapur® 200, JRS Pharma Rosenberg, Germany). The particle size distributions of all materials are presented in Fig. 1 and characteristic values in Table 3:

Except for the microcrystalline celluloses, all materials were mixed with magnesium stearate (+1 wt%) (LIGAMED MF-2-V, Peter Greven GmbH & Co. KG, Bad Münstereifel, Germany) in a bin blender for 15 min at 30 rpm for lubrication. The variety of materials was chosen to provide a broad range of flowability and permeability (Fig. 2).

3. Methods

3.1. Powder characterization

Particle size measurements were carried out by using laser diffraction with a dry dispersion unit applying a dispersion pressure of 0.5 bar (Mastersizer 3000 and Aero S, Malvern Panalytical, Malvern, United Kingdom). The powder flow behavior was measured using the granulate

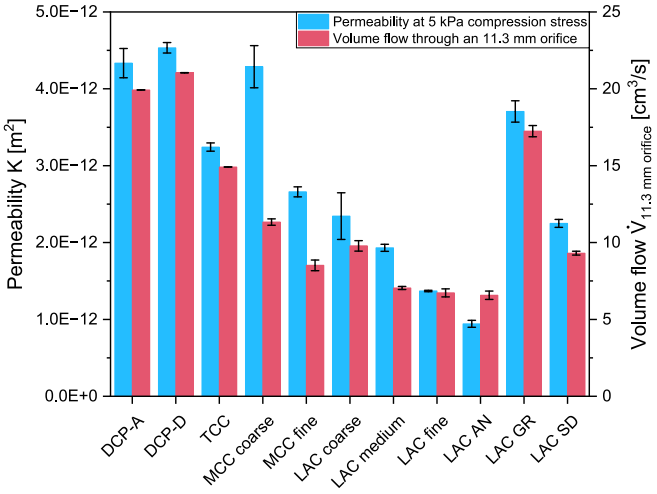


Fig. 2. Permeabilities and volume flows through an orifice for all materials (without MgSt) investigated.

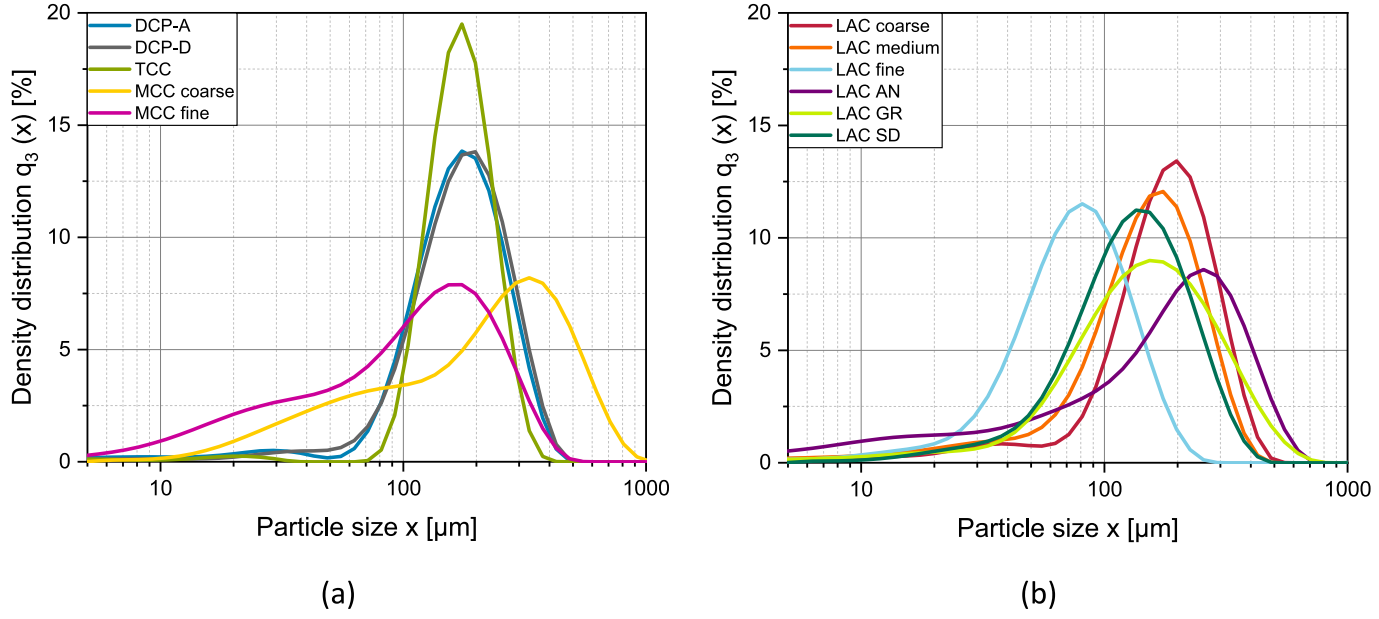


Fig. 1. Particle size distributions of DCP-A, DCP-D, TCC, MCC coarse and fine (a) and the lactose grades (b), measured by laser diffraction with a dry dispersion unit using dispersion pressure of 0.5 bar.

flow tester GTB (Erweka GmbH, Langen, Germany) with a 11.3 mm orifice without the stirring unit. A hopper was filled with powder and the opening was automatically opened to initiate the powder flow. The flow rate was determined by means of time and weight measurement. The specific flow duration (s/100 g) was converted into a volume flow (ml/s) using the bulk densities (Table 3). The powder rheometer MCR 302 (Anton Paar, Graz, Austria) was used to measure the permeability of the powders. Therefore, 80 ml of powder was fluidized by an air flow of 5 l/min for 200 s and additional rotation by a two-blade stirrer with 5 rpm. After fluidization, the powder was consolidated by an air-permeable metal plate with stresses from 1 to 20 kPa before two different air flows of 0.5 and 1 l/min were applied. The resulting differential pressures were measured and the permeability was calculated based on Darcy's law [19,20]. The bulk and tapped densities were measured as described in Ph. Eur. 2.9.34. Therefore, a volumetric analyzer was used (Erich Tschacher Laboratoriumsbedarf, Bielefeld, Germany). For all methods of powder characterization, three measurements were conducted and mean values were calculated.

3.2. Die filling experiments

Two different rotary tablet presses were used to investigate the die filling under suction. The production-scale tablet presses (X 3 and XL 400, KORSCH AG, Berlin, Germany) differ in their pitch diameter and their number of tools (Table 1). All tablet presses were equipped with round and flat-faced 9 mm EU-D tools. The experiments were carried out using single-paddle feed frames, which have the advantage that the lower punch is pulled down completely under the paddle wheel. Investigating the filling behavior using two-paddle feed frames entails the challenge that the pull-down partly takes place under a space between the paddle wheels with probably different powder bed conditions. Additionally, the dosing wheel usually rotates in an opposite direction to the filling wheel, making it more difficult to clearly attribute the filling performance to specific effects so the focus of this study is on single-paddle feed frames. Due to the steeper fill cams and the resulting higher wear propensity on single-paddle compared with two-paddle feed frames, the turret speed is limited to a maximum of 60 rpm (Table 1). Furthermore, both the clockwise (cw) and counterclockwise (ccw) rotational direction of the filling wheel were investigated.

Due to limited capacities and availabilities, not all experiments could be carried out using the same paddle wheel geometry, meaning that the spoke height of the paddle wheels varied (Table 2). The different paddle wheel heights used for the respective experiments, in combination with the different geometries of the feed frames lead to varying clearances between the lid of the feed frames and the paddle wheels (Table 2; Fig. 1 in the supplementary material). Nevertheless, paddle wheels with straight, rectangular spokes were used. Methods and results for lab scale experiments (XL 100) on gravity filling were previously published and are used as comparison in this study [12].

The filling and dosing heights were kept constant at $h_F = 18$ mm and $h_D = 17.5$ mm. These filling heights are significantly higher than those normally used for industrial tablet production, but increase the probability of incompletely filled dies. This is important, as completely filled dies do not add new information for model development.

All tableting experiments were carried out by gradually changing the turret and paddle speeds by 10 rpm and all possible combinations were investigated. Samples were taken 30 s after each adjustment of a new parameter combination. 20 tablets were weighed using the automated tablet tester MultiCheck 6 (Erweka GmbH, Langen, Germany) or the semi-automatic tester LAB.line P4 (Kraemer Elektronik GmbH, Darmstadt, Germany). Means and standard deviations were calculated (±1 σ range is shown in the figures).

Table 1
Process parameters for the die filling experiments. *Method and results described in [12].

	XL 100*	X 3	XL 400
Setup	Gravity filling	Suction filling	Suction filling
Number of dies	4	23	29
Pitch diameter [mm]	118	325	410
Turret speed [rpm]	30–60	40–60	40–60
Paddle speed [rpm]	5–60	10–60, 100	10–60, 100

Table 2
Used paddle wheel heights and resulting clearances.

Rotary tablet press	Material	Paddle wheel height	Clearance between filling wheel and lid
X 3	DCP-A, MCC coarse, MCC fine	5.5 mm	7.75 mm
	DCP-D, TCC, LAC coarse, LAC medium, LAC fine, LAC AN, LAC GR, LAC SD	8.0 mm	5.25 mm
XL 400	DCP-A, DCP-D, TCC, MCC coarse, MCC fine	5.5 mm	3.3 mm
	LAC coarse, LAC medium, LAC fine, LAC AN, LAC GR, LAC SD	5.9 mm	2.9 mm

3.3. Calculation of the volume flow into the dies

Based on the developed approach to model gravity-driven die filling in previous studies, the powder volume flow rate $\dot{V}$ into the dies was calculated based on the filling time t_{fill}. The filling time results from the ratio of filling angle α_{fill} and the turret speed (expressed as its rotation frequency 1/min) n_t:

$$t_{fill} = \frac{\alpha_{fill}}{n_t \cdot 360^\circ} \tag{2}$$

Following, the mean tablet weight $\overline{m}_T$is divided by the filling time and the respective bulk density ρ_b of the material to determine the volume flow rate $\dot{V}$:

$$\dot{V} = \frac{\overline{m}_T}{t_{fill} \cdot \rho_b} \tag{3}$$

4. Results and discussion

This study aims to develop a model to predict the suction-driven die filling on production scale rotary presses only based on independently measured powder properties. In this context, the critical paddle speed to fill the dies completely is the criterion used to validate the model. Such a model allows the prediction of process parameters based on few experiments of powder characterization for single-paddle feed frames. In order to achieve this goal, it is important to deepen the understanding of the impact of suction filling on rotary tablet presses in general and on the impact of process parameters and material properties in specific.

4.1. Results of the powder characterization

The investigated materials differ in their particle size distributions and their bulk and tapped density (Fig. 1 and (Table 3). The powder properties flowability and permeability, which are particularly relevant for die filling, also differ substantially for the different materials. Materials with a relatively broad particle size distribution, such as MCC fine or LAC AN, have relatively low permeability and poorer flowability. In contrast, materials with a relatively narrow particle size distribution, such as DCP-A or DCP-D, which also have a comparatively high bulk density, exhibit good flowability. In comparison, their permeability is also relatively high. The selection of materials covers a wide range of flowability and permeability in order to enable the broad applicability of the model.

Table 3
Powder properties of all materials investigated.

Material	x_{10} [µm]	x_{50} [µm]	x_{90} [µm]	Span = (x_{90} - x_{10}) / x_{50} [−]	Bulk density ρ_b [g/cm³]	Tapped density ρ_t [g/cm³]
DCP-A	85.7	168	280	1.16	0.75	0.83
DCP-D	89.1	175	290	1.15	0.90	0.97
TCC	112	169	247	0.80	0.63	0.69
MCC coarse	41.2	219	499	2.09	0.37	0.51
MCC fine	19.4	110	254	2.13	0.37	0.52
LAC coarse	73.3	178	299	1.27	0.82	0.89
LAC medium	49.6	149	263	1.43	0.81	0.90
LAC fine	31.8	74.8	140	1.45	0.72	0.85
LAC AN	14.8	172	378	2.11	0.72	0.91
LAC GR	55.7	148	328	1.84	0.61	0.66
LAC SD	56.5	130	243	1.43	0.72	0.79

4.2. Volume flow rates for gravity and suction filling

In a first step, volume flow rates into dies (comp. eq. (3)) for gravity and suction filling are compared for DCP-A and LAC coarse exemplarily, which exhibit different flow properties and permeabilities (Fig. 3). DCP-A presents a higher flow through an orifice and has a higher permeability than LAC coarse. Descriptions of the methods to obtain data for gravity filling can be found in a previous paper [13].

Necessary volume flow rates to fill the dies completely $\dot{V}_{comp}$ rise with increasing turret speed and are indicated by dashed lines, parallel to the x-axis. $\dot{V}_{comp}$ can be calculated based on the dosing volume V_D, resulting from the dosing height and the die diameter, and the filling time t_{fill} (Eq. (2)) and are accordingly turret speed-dependent:

$$\dot{V}_{comp} = \frac{V_D}{t_{fill}} = \frac{V_D \cdot n_t \cdot 360^{\circ}}{\alpha_{fill}} \tag{4}$$

As already presented in former studies with focus on gravity filling, a general increase in volume flow rate over paddle speed can also be observed for suction filling [12,13]. However, for both materials, considerably higher values but smaller slopes compared to gravity filling are observed. During gravity filling, DCP-A exhibits higher volume flow rates than LAC coarse, which can be traced back to coarser particles, lower cohesion and higher permeability, and thus better flow properties for DCP-A. This superior behavior changes for suction filling, as volume flow rates are considerably higher for LAC coarse, especially with increasing turret speed. This finding aligns perfectly with the literature on test rigs, which presents an increasing impact of a differential pressure on the filled powder volume for materials with decreasing permeability [6,7,14], confirming its phenomenological transferability to rotary presses in production scale.

Generally, the linear course, which was identified previously for gravity filling, is also apparent for suction filling for sufficiently high paddle speeds and thus, can be interpreted as fundamental relationship between powder flow into the dies and this process parameter. However, at low paddle speeds, volume flow rates decline and therefore do not follow the linear trend. Here, the powder demand for die filling cannot be met due to the slow rotation of the paddle wheel and the associated slow transport of the powder into the filling area. These observations present the boundaries of the directly proportional correlation between volume flow rate and paddle speed in the investigated system. The paddle and feed frame height, the number of spokes and the number of dies can have an impact on the deliverable volume flow rate on the machine side. Since the investigation of these limits is not the focus of this study, data points at low paddle speeds are not taken into account for model fitting, as they represent an unstable process with high deviations ($\sigma > 5$ %).

For further analysis, the linearly rising volume flow rates are fitted by the linear eq. (5) to investigate the axis intercept, called reference volume flow during suction filling $\dot{V}_{0,s}$, and the slope, called rotation-specific fill volume during suction filling $V_{r,s}$ (Fig. 4, eq. (5)) in accordance to the respective parameters implemented for gravity filling [12]. Both parameters represent the material-specific sensitivity to the suction filling process.

$$\dot{V}_s = V_{r,s} \bullet n_p + \dot{V}_{0,s} \tag{5}$$

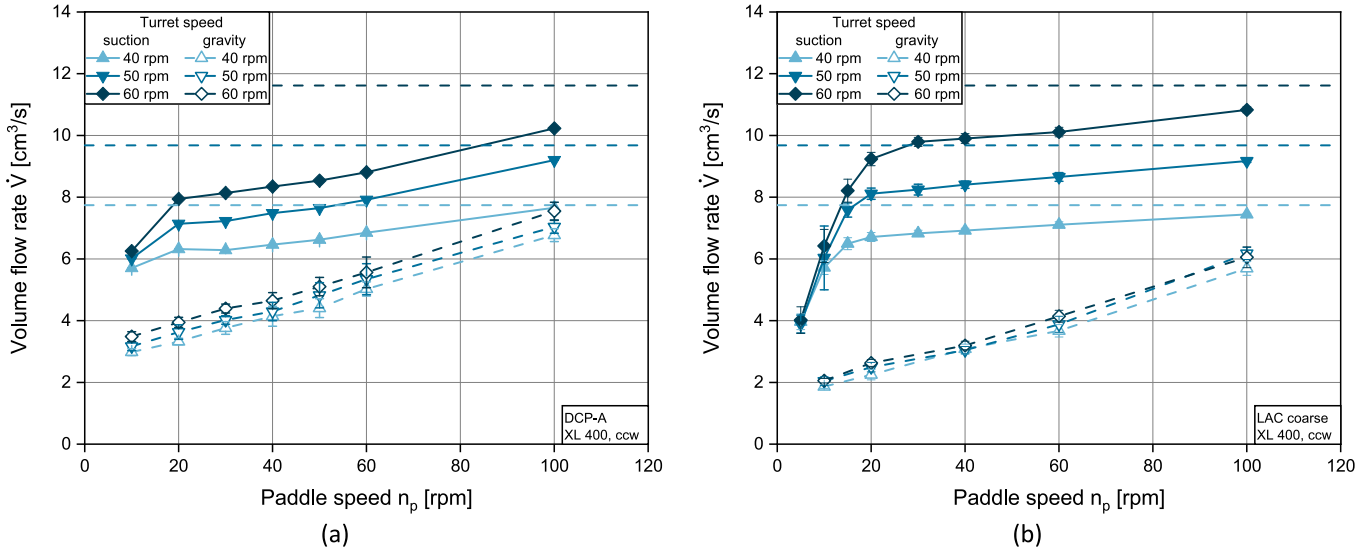


Fig. 3. Volume flow rates into the dies during gravity and suction filling for DCP-A (a) and MCC fine (b) for XL 400 ccw (horizontal lines display volume flow rate to fill dies completely $\dot{V}_{comp}$ according to eq. (4)).

The applied fit cannot only be used to describe the volume flow rate in dependency on the paddle speed, but also to identify the minimal paddle speed to ensure complete die filling, $n_{p,crit}$. It can be calculated based on the intersection with $\dot{V}_{comp}$ for the desired turret speed (eq. (4)) and eq. (5):

$$n_{p,crit} = \left(\frac{V_D \cdot n_t \cdot 360^{\circ}}{\alpha_{fill}} - \dot{V}_{0,s}\right) \cdot \frac{1}{V_{r,s}} \tag{6}$$

4.3. Theoretical calculation of the differential pressures

A fundamental understanding of the relations between lower punch movement and the evolving differential pressure enables a physically based description of the impact of process parameters. In general, it is known that the pull-down of the lower punch below the powder bed provokes the development of a differential pressure inside the die [2]. With increasing turret speed, the pull-down velocity increases while the filling time decreases. In the following, an approach is presented to estimate the differential pressure inside the die, which is assumed to have a direct impact on the filled powder weight. For this approach, some assumptions are made to enable the calculation of a differential pressure:

1) the volume expansion inside the die and the air flow through the powder bed are considered
2) the air is considered as ideal gas
3) the particle volume entering the die is not taken into account
4) air flow through the gap between die and lower punch is neglected

Generally, this approach provides an estimation of the potential maximum differential pressure against ambient conditions. Accordingly, it should provide reasonable scaling results for the combination of punch movement (process parameters) and powder permeability (material properties). However, due to the simplifying assumptions the actually occurring differential pressures will be smaller, but should scale in a comparable manner to the discussed results here.

The main impact factors are the expansion of the free volume inside the die, connected to the pull-down velocity, and the airflow through the powder bed above the die (assumption 1). Both can be calculated from known correlations. The lower punch movement with the pull-down velocity $dh/dt = v_p$ creates in connection with the cross-sectional area A_{die} a rising volume inside the die over time dV_{die}/dt (Fig. 5):

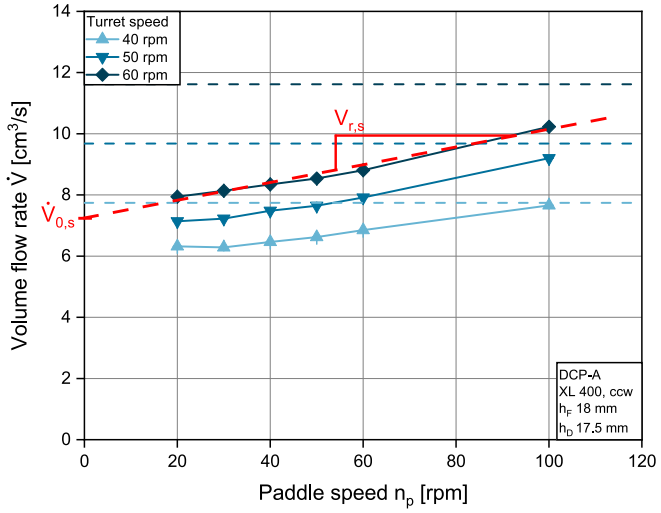


Fig. 4. Illustration of data fitting and extraction of the reference volume flow $\dot{V}_{0,s}$ and the rotation-specific fill volume $V_{r,s}$ (horizontal lines display volume flow rate to fill dies completely $\dot{V}_{comp}$ according to eq. (4)).

$$\frac{dV_{die}}{dt} = A_{die} \cdot \frac{dh}{dt} = A_{die} \cdot v_p \tag{7}$$

In general, the pull-down of the lower punch is not completely linear as an initial vertical acceleration and deceleration has to take place. Thus, fill cams exhibit usually a curved, s-shaped profile (Fig. 6 a).

However, as only the beginning and the end of the pull-down are considerably curved, for a decent filling depth, a mean pull-down velocity v_p can be identified, which describes the actual velocity with a sufficiently high accuracy. Therefore, the filling height h_F is divided by the time the punch is effectively pulled downwards by the fill cam t_p:

$$v_p = \frac{h_F}{t_p} = \frac{h_F \cdot 360^{\circ} \cdot n_t}{\gamma_p} \tag{8}$$

The pull-down time t_p is based on the turret speed n_t and the suction angle γ_p, which was identified based on technical drawings of the fill cams used on the X 3 and XL 400 (illustration in Fig. 6 b).

Under the assumption that the air behaves like an ideal gas (assumption 2), the ideal gas law is applied to correlate the changing volume over time dV_{die}/dt with the time dependent pressure $p(t)$ and the amount of air molecules n_{air}, which also change over time due to the inflowing air through the powder (Fig. 5):

$$p(t) \cdot \frac{dV_{die}}{dt} = \frac{dn}{dt} \cdot R \cdot T \tag{9}$$

R is the general gas constant and T the temperature in Kelvin. The ideal gas law is applicable for gases without any molecular interaction, with a negligible molecule volume and without rotation or vibration. At a normal state (0 °C, 101,325 Pa), air behaves almost as an ideal gas and as filling experiments are conducted at ambient conditions (20 °C, ambient pressure) the difference in temperature is neglected and the ideal gas law is applied. Nevertheless, the simplified assumption can lead to deviations, as the exact state of the air in this case can only be estimated, but not measured.

In order to find an explicit description for $p(t)$, time dependent variables have to be replaced. The time dependent change of the free volume inside the die dV_{die}/dt can be expressed by the product of the dies' cross-sectional area A_{die} and the pull-down velocity of the lower punch v_p.

The changing amount of molecules dn/dt can be expressed by considering the airflow through the powder Q_{air}, its density ρ_{air} and its molecular weight M_{air}:

$$\frac{dn}{dt} = \frac{Q_{air} \cdot \rho_{air}}{M_{air}} \tag{10}$$

The powder bed above the die is considered to be an air-permeable frit that allows air to pass through without powder entering the die or changing its permeability. The airflow through the powder Q_{air} can be

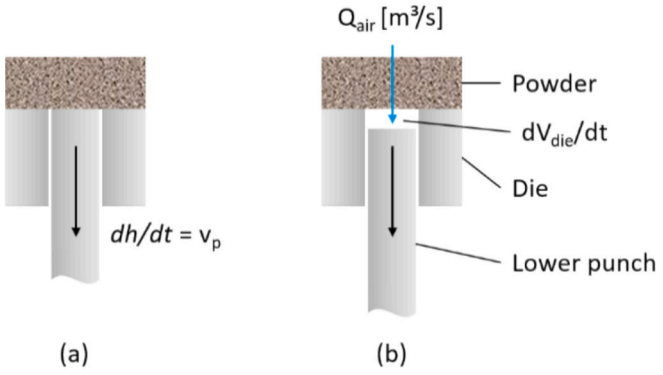


Fig. 5. Schematic illustration of the punch movement below the powder bed (a) and the resulting airflow through the powder (b).

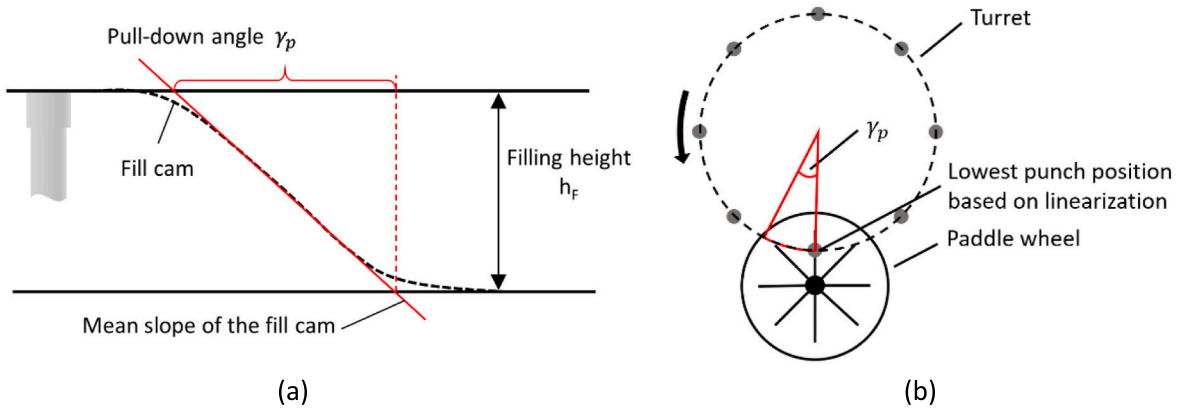


Fig. 6. Schematic illustration of the fill cam (dashed line) and the estimated mean slope (red line) of it in a side view (a) and illustration of the pull-down angle in a top view (b).

calculated based on Darcy's law for permeable materials, which exhibit the permeability K:

$$Q_{air} = \frac{K \cdot A_{die} \cdot (p_0 - p(t))}{\eta_{air} \cdot L} \tag{11}$$

Further, the dies' cross-sectional area A_{die}, the dynamic viscosity η_{air}, the flown through length L, which equals the powder bed height in the feed frame (paddle height used), and the difference between ambient pressure p_0 and actual pressure inside the die $p(t)$, i.e. *the differential pressure,* are considered.

Inserting these expressions in eq. (10), the following equation results. Here, $p(t)$ is replaced by p_{stat} as no parameter is considered to be time dependent anymore. In addition, the parameter c is used, which combines all constants for a specific material and a specific powder bed height, thereby simplifying the equation.

$$p_{stat} \cdot A_{die} \cdot v_p = \frac{K \cdot \rho_{air} \cdot A_{die} \cdot (p_0 - p_{stat})}{\eta_{air} \cdot M_{air} \cdot L} \cdot RT \tag{12}$$

$$p_{stat} = \frac{\frac{c}{v_p} \cdot p_0}{\frac{c}{v_p} + 1} = \frac{c \cdot p_0}{c + v_p} \tag{13}$$

$$c = \frac{K \cdot \rho_{air}}{\eta_{air} \cdot M_{air} \cdot L} \cdot R \cdot T \tag{14}$$

The differential pressure Δp against ambient conditions p_0 can be calculated as followed by this broken rational equation:

$$\Delta p = p_0 - p_{stat} = p_0 - \frac{c \cdot p_0}{c + v_p} = p_0 \cdot \left(1 - \frac{c}{c + v_p}\right) = p_0 \cdot \frac{v_p}{c + v_p} \tag{15}$$

Stationary conditions imply a constant pull-down velocity and a constant airflow through the powder bed, while no particles enter the die. It can be assumed, that these conditions are only approximately met at the beginning of the pull-down, so the calculation describes the potential maximum differential pressure.

As the volume of the particles inside the die over time cannot be assessed, it cannot be included into calculations (assumptions 3). In a simulative study, Wu and Guo presented a considerable differential pressure for a constantly moving punch although particles enter the die [15]. Particles reduce the overall free volume increase and, probably, increase the permeability and lower the effective powder bed height above the die. Therefore, it can be assumed that calculated differential pressures (eq. (15)) overestimate actual conditions in real processes. However, the powder porosity is high during flow into the die, so deviations should be in a reasonable extent, which means this approach allows comparing different pull-down velocities and materials.

The airflow through the gap between the lower punch and die wall is also neglected, as it is assumed to be small in relation to the airflow through the powder bed (assumption 4). This favors an additional overestimation of the differential pressure.

The calculated differential pressure (eq. (15)) also differ for the rotary presses X 3 and XL 400 depending on the assumed powder bed height of 5.5 mm and 8 mm, which reflects the range of the used paddle heights (Table 2). The higher the powder bed height, the higher the expected differential pressure (comp. Fig. 2 in the supplementary material). With $R^2 > 0.999$, the relationship between the pull-down velocity and the differential pressure can be approximated linearly for both powder bed height considered. As the powder bed height has a considerable impact on the developing differential pressure, it should be investigated in more depth in the future. Although the rotary presses have considerably different pitch diameters, pull-down velocities are in a comparable range between 100 and 400 mm/s due to the design of the fill cams.

The calculated differential pressure (eq. (15)) strongly varies for the materials used, based on their diverging permeability (Fig. 7). Higher permeabilities correspond to higher values of c and result in lower values for Δp. For DCP-A and other materials with values of $c > 2$ an almost linear increase in Δp over v_p can be identified. However, with decreasing permeability and, thus with c-values closer to 1, the deviation from a linear course rises (based on its influence in eq.(15)), so for LAC AN a more curved graph is visible.

4.4. Model of die filling for single-paddle feed frames

The main goal of the model is to calculate the volume flow rates into the dies based on material properties and process parameters, so it reproduces the linear courses of the actual volume flow rates as accurately as possible (comp. Fig. 4). Therefore, the axis intercept, called *reference volume flow*, and the slope, called *rotation-specific fill volume*, of the model equation are discussed separately in the following. In order to make the model as generally applicable as possible, all investigated powders and the two tablet presses are included into model development.

Only data from experiments with incompletely filled dies is used for modeling, as the model eq. (5) describes a direct relationship between the powder volume flow rates and the paddle speed. With suction filling, some materials show complete filling regardless of the paddle speed, so no linear increase can be identified. Additionally, data points indicating densification during die filling are excluded for modeling, as the theoretical maximum filling volume is exceeded. The densification of the powder within the die beyond the bulk density follows other mechanisms than the suction filling addressed in this model. Accordingly, the phenomenon of densification within completely filled dies is likely correlated to a complex interaction between the paddle speed and the

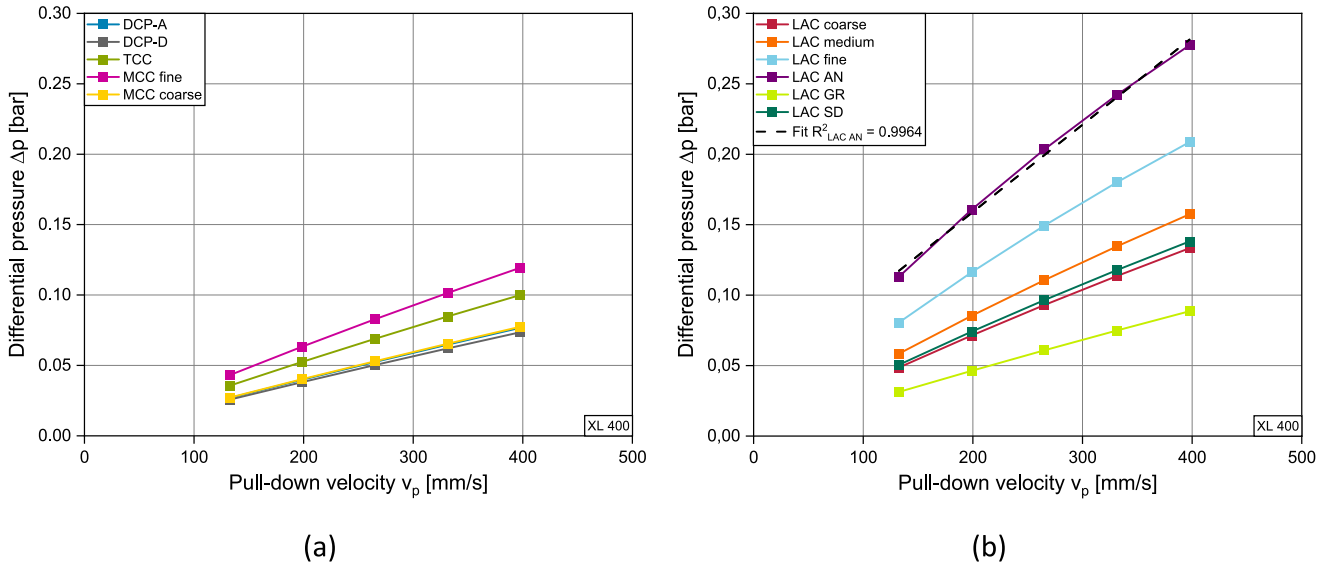


Fig. 7. Differential pressure calculated for DCP-A, DCP-D, TCC, MCC coarse, MCC fine (a) and LAC coarse, medium, fine, AN, GR, SD (b) on the XL 400 using eq. (15).

compressibility of the powder, but not addressed in this study. Thus, some data sets are excluded from model development (Table 4).

4.4.1. Model of the reference volume flow $\dot{V}_{0,s}$ as a function of the differential pressure

The reference volume flow $\dot{V}_{0,s}$ is the axis intercept of the linearly fitted volume flow rates over paddle speed (comp. Fig. 4) and it represents a theoretical volume flow into the die without the forced feeding effect of the paddle wheel. Fig. 4 shows that $\dot{V}_{0,s}$ shifts to higher values for increasing turret speed. Considering the differential pressure Δp as the main difference between gravity and suction filling, it is a reasonable parameter to use for model development. Calculating Δp enables the comparison of different rotary presses by taking the turret speed and pull-down angle of the fill cams into account. In the literature, it was already shown that higher powder volume flow rates through an orifice from a bin can be achieved at higher differential pressures [17].

Fig. 8 shows exemplarily the relationship between the reference volume flow $\dot{V}_{0,s}$ and the differential pressure Δp for DCP-A and LAC coarse, which differ in terms of flowability and permeability (comp. Fig. 2). Data points of both rotary presses (X 3, XL 400) and both directions of rotation can be described by the following equation:

$$\dot{V}_{0,s}(\Delta p) = a_s \cdot \Delta p + b_s \tag{16}$$

The reference volume flow for gravity filling $\dot{V}_0$ ($\Delta p = 0$ Pa) of the XL 100, originating from a previous study [12], matches to the extrapolation of the suction filling data of X 3 and XL 400 well. This is in line with the fact, that the XL 100 reflects the lower boundary of the model because its missing punch pull-down (gravity filling) yields a differential pressure of $\Delta p = 0$ Pa. Consequently, it is included in the model development, which makes the fit less susceptible to deviations within the data of the large rotary presses grouping in specific ranges of differential pressure, respectively. Regarding the applicability of the model, this facilitates that the model can potentially also be transferred to tablet presses with different fill cams or turret diameters or speeds and that the resulting negative pressure can be seen as the main influencing factor. Reference volume flows from gravity filling experiments on the X 3 and XL 400 cannot be integrated in these graphs, as an impact of a dynamic differential pressure due to the Bernoulli effect of quickly moving dies was observed [13].

Table 4
Experiments excluded from model development.

Rotary tablet press	Direction of rotation	Material
X 3	cw	MCC fine
		LAC fine
	ccw	MCC fine

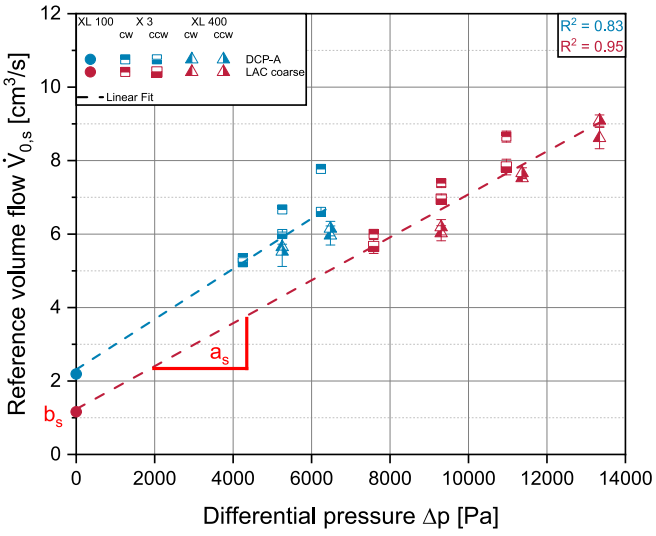


Fig. 8. Correlation between the differential pressure and the reference volume flow for DCP-A and LAC coarse.

The X 3 presents higher $\dot{V}_{0,s}$ values than XL 400 (Fig. 8), which might be caused by the higher clearance between the paddle wheels and the lid of the feed frame at the X 3 (see Table 2). A higher clearance may result in a higher powder bed. On one hand, this increases the amount of available powder for die filling. On the other hand, a higher powder bed enhances the resistance against an air flow and results in a lower c-values (comp. Eq. (14)). For the same pull-down velocity, this leads to higher differential pressures and thus, higher $\dot{V}_{0,s}$ (comp. Fig. 2 in the supplementary material). Different paddle heights as well as the direction of rotation did not show a systematic impact on $\dot{V}_{0,s}$. In the future, the impact of paddle height, clearance and direction of rotation should be investigated in more depth.

$\dot{V}_{0,s}$ presents a higher intercept and a higher slope over Δp for DCP-A compared to LAC coarse, which can be explained by different powder

properties (Fig. 8). The eleven powders used cover a broad range of flowability and permeability. Following, the parameters a_s and b_s are correlated with the permeability and the volume flow through an orifice in order to develop an approach with broad applicability.

The slope a_s of the material-specific linear fits for $\dot{V}_{0,s}$ over Δp (eq. (16)) can be interpreted as susceptibility to the differential pressure. This parameter correlates linearly with the permeability, which describes the resistance against an air flow through the powder. Mills et al. already suggested a strong impact of the powders' permeabilities and Zakhvatayeva et al. discussed a lower suction efficiency in correlation with a higher permeability [6,14]. The correlation presents a higher slope a_s for powders with high permeability. This result seems contra-intuitive as fine powders benefit more from a given differential pressure. However, even at low turret speeds powders with low permeability enable the development of a much higher differential pressure (comp. (15), Fig. 7). LAC AN, for example, has the lowest permeability of the investigated materials, which helps to build up a higher differential pressure. As a result, although values for $\dot{V}_{0,s}$ might be higher for low permeabilities, Δp is much higher at a given turret speed, which leads to lower slopes in a $\dot{V}_{0,s}$ - Δp plot.

A linear relationship of a_s and K can be identified (Eq. (17)):

$$a_s = a_1 \cdot K + a_2 \tag{17}$$

With the constants $a_1 = 1.174 \bullet 10^8 \frac{cm^3}{s \bullet Pa \bullet m^2}$ and $a_2 = 2.812 \bullet 10^{-4} \frac{cm^3}{s \bullet Pa}$.

Parameter b_s represents a theoretical reference volume flow without a differential pressure, like during gravity filling. A good agreement was found between the parameter b_s and the volume flow through an orifice $\dot{V}_{Orifice}$ for all eleven materials:

$$b_s = b_1 \cdot \dot{V}_{Orifice} + b_2 \tag{18}$$

With the constants $b_1 = 0.011$ and $b_2 = 0.071 \frac{cm^3}{s}$.

Correlating the reference volume flows to powder properties is the first step towards the transferability of the model to any powders. There are many methods to characterize the flowability of a powder, but the volume flow through an orifice seems to be the most similar to the powder flow into a die in terms of particle flow. In fact, the orifice diameter is only 2.3 mm larger than the die diameter used (11.3 mm) and it was the smallest diameter at which all powders were still able to flow. In previous studies on gravity filling, such as those reported by Schomberg et al., the modeling of the reference volume flow was based on the bulk density and not on the volume flow through an orifice [12,13]. This difference underlines that powder flow properties can be described in multiple ways and that different filling mechanisms may require different characterization methods to achieve the most reliable predictions. The flow properties were not only used to model the volume flows but also to model the mass variations during die filling [21,22]. It was shown that powders with poorer flow properties lead to more inconsistent die filling. However, the investigation of the variations during die filling is not the aim of this study.

Finally, the following mathematical relationship can be used to describe the reference volume flow $\dot{V}_{0,s}$ based on the differential pressure Δp and the material properties permeability K and volume flow through an orifice $\dot{V}_{Orifice}$:

$$\dot{V}_{0,s}(\Delta p) = (a_1 \cdot K + a_2) \cdot \Delta p + b_1 \cdot \dot{V}_{Orifice} + b_2 \tag{19}$$

The various mathematical approximations and the resulting deviations reduce the predictive quality of the model. These deviations can have various reasons, including the effects that are still not understood properly, such as the direction of rotation of the paddle wheels, the geometry of the feed frames or the dynamic conditions of the powder inside the feed frame such as powder bed height and compression state. The accuracy of the linear approximations is also influenced by the results of the powder characterization, whereas cohesive materials can show inconsistent powder flow making it difficult to reproducibly determine the flowability.

A noticeable deviation can be observed for LAC GR (comp. Fig. 9). Compared to the other lactoses, LAC GR has a good flowability and a high permeability (comp. Fig. 2). Nevertheless, it has values for a_s and b_s that are not adequately captured by the linear fits. A possible explanation could give the particle structure. LAC GR is a granulated grade, consisting of several primary particles. During the feed frame passage, abrasion and particle breakage is possible, which could result in an overall finer particle size, which would be connected to a change in permeability and volume flow through an orifice.

The consequences of model simplifications are discussed in Section 3.5. There, critical paddle speeds for complete die filling are calculated using the model and compared with experimentally determined values.

4.4.2. *Model of the rotation-specific fill volume $V_{r,s}$ as a function of the differential pressure*

The rotation-specific fill volume $V_{r,s}$ describes the increase in volume flow rate $\dot{V}_s$ with rising paddle speed n_p (comp. Fig. 4). While it is reasonable that the volume flow rates grow for higher paddle speeds, which results in a higher amount of powder per time unit in the filling area, it is worth to mention that this relation also depends on the turret speed. The slope $V_{r,s}$ increases with rising turret speed, as already identified for gravity filling on production scale rotary presses [13]. In order to consider the scale of the rotary press and the main difference between gravity-driven and suction-driven die filling, the differential pressure Δp is again used as scaling parameter.

With increasing differential pressure, $V_{r,s}$ rises, as exemplarily presented for DCP-A and LAC coarse (Fig. 10). In general, the increase of $V_{r,s}$, observable for all materials, could be explained by process conditions: A higher production speed reduces the powder hold-up inside the feed frame, as more dies per time unit are filled with a higher powder volume due to the increased suction. Thus, the demand of powder is increased and the die filling benefits from a constant "supply" due to the rotation of the paddle wheel.

The rotation-specific fill volume $V_{r,s}$ is similar for cw and ccw for the rotary press X 3. However, for the XL 400 diverging values for cw and ccw can be observed for both materials presented (Fig. 10). Thus, it is subject to some uncertainties, which are not understood in depth, yet. $V_{r,s}$ is systematically lower for the X 3 compared to the XL 400 for all materials. This suggests the effect of a difference in the construction of the rotary presses, especially of the feed frame. A measurable difference is the clearance between the paddle wheel and the lid of the feed frame, which is considerably smaller for the XL 400 than for the X 3 (Table 2). Following the discussion on the hold-up, it might be higher for the X 3, as more powder can initially enter the feed frame and can be transported between the spokes and above, called waving, which was observed in the literature [23,24]. As discussed in Section 3.4.1, more powder might be initially available for die filling at a given paddle speed, resulting in slightly higher values for the axis intercept $\dot{V}_{0,s}$. Consequently, an increase in paddle speed has a slightly lower effect on die filling compared to the XL 400. The XL 400 benefits more from the higher paddle speed, and thus, higher amount of powder in the filling area, due to the potentially lower initial powder bed height. Additionally, the lower hold-up compared to the X 3 might enhance the particle movement, contributing to higher values of $V_{r,s}$ (Fig. 10) on the XL 400. However, these hypotheses have to be proved in the future, e.g., by analyzing the powder bed height, hold-up in the feed frame and the actual powder density. Interestingly, the different paddle wheel heights used for DCP-A (5.5 mm) and LAC coarse (X 3: 8 mm, XL 400: 5.9 mm, Table 2) do not seem to have a strong impact on $V_{r,s}$. Deviations between the rotary presses are in the same range for both, which rules out the potential advantage of a higher paddle height. Nevertheless, the geometry of the paddle wheels is generally considered to be very relevant and influential to the die filling depending on aspects such as the number, cross-sectional geometry or curvature of the spokes [25].

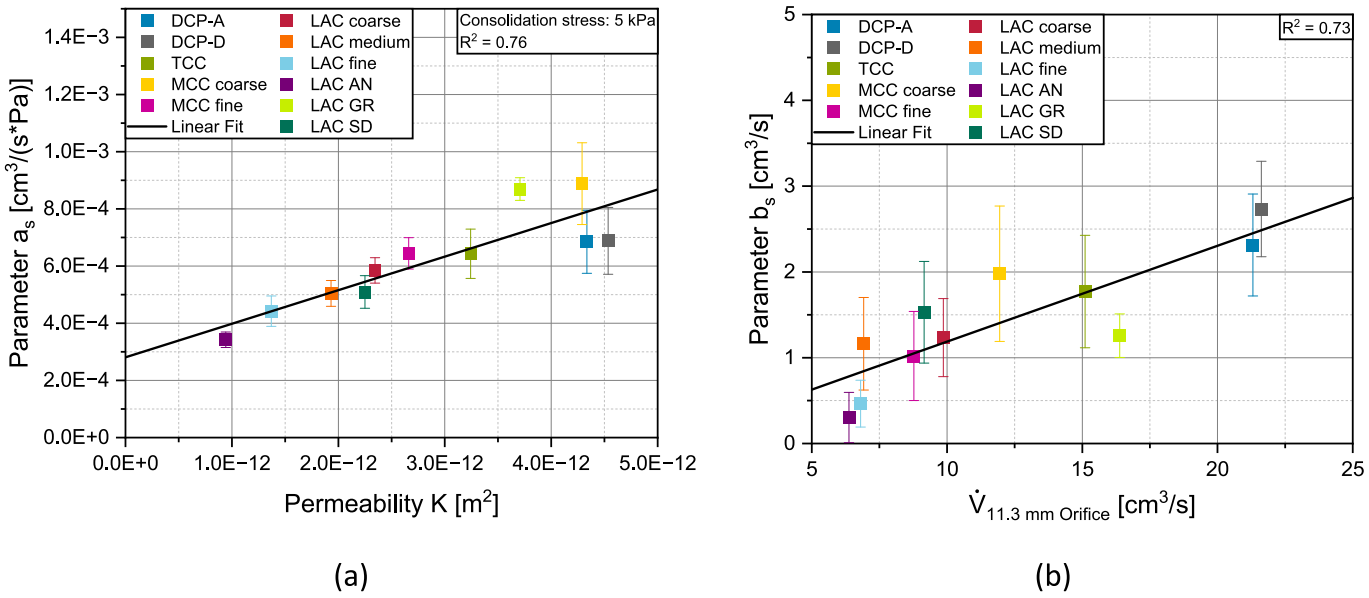


Fig. 9. Correlations between the parameters a_s with the permeability K (a) and b_s with the flow through an orifice (b) for all investigated powders.

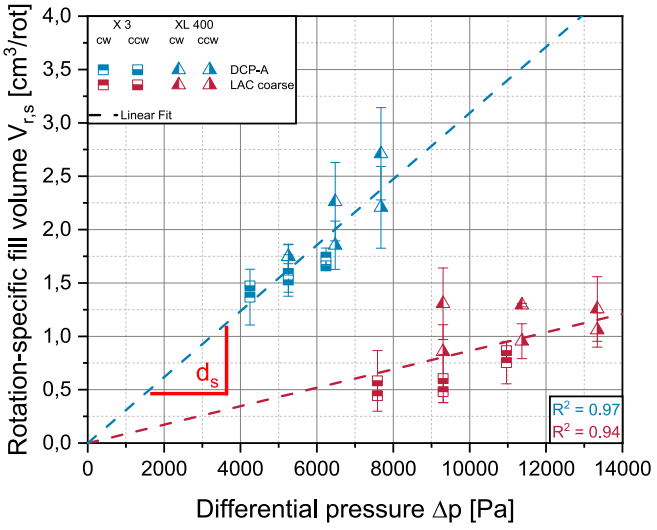


Fig. 10. Rotation-specific fill volume over differential pressure for DCP-A and LAC coarse.

In order to model the change in rotation-specific fill volume over differential pressure, a linear function is used [13]:

$$V_{r,s} = d_s \cdot \Delta p \tag{20}$$

For suction filling, values for $V_{r,s}$ are low and converge to zero with decreasing differential pressure (Fig. 10). This can be explained by the reduced powder demand with decreasing differential pressure and therefore the reduced dependency on the paddle wheel rotation. That is why an approximation without an axis intercept is used to reflect the data. $V_{r,s}$ for all eleven materials used in this study is investigated and approximated by eq. (20). The parameter d_s is extracted to find a correlation with a measurable powder property.

The parameter d_s presents a linear correlation with the permeability of the powder – it rises with increasing permeability (Fig. 11). Consequently, the rotation-specific fill volume rises over Δp stronger for materials with high permeability. This means that the paddle movement has a higher impact on the volume flow rate for more permeable powders, like DCP-A. Having a look at the die filling performance, these powders (e.g. DCP-A) possess lower volume flow rates compared to powders with lower permeabilities (e.g. LAC coarse) (Fig. 3). As already discussed, this can be explained by the lower estimated differential pressure. As Mills et al. already suspected and Baserinia et al. presented by measurements, a gravity filling phase after the suction filling is expected [6,7]. More permeable powders may have a less distinct volume flow rate during suction filling and benefit more from the forced feeding effect, reflected by the factor d_s. Although several powder properties can be responsible for better die filling behavior using rotating paddle wheels, the permeability presents the highest coefficient of correlation with d_s (Fig. 11: $R^2 = 0.87$), resulting in the following equation:

$$d_s = d_1 \cdot K - d_2 \tag{21}$$

With the constants $d_1 = 6.567 \bullet 10^7 \frac{cm^3}{rot \bullet Pa \bullet m^2}$ and $d_2 = 5.271 \bullet 10^7 \frac{cm^3}{rot \bullet Pa}$.

Nevertheless, MCC coarse and LAC GR deviate slightly from the correlation. The parameter d_s is overestimated for these materials, which will result in an overestimated value for $V_{r,s}$. Since the accuracy of the modeled volume flow rate strongly depends on $V_{r,s}$, deviations are

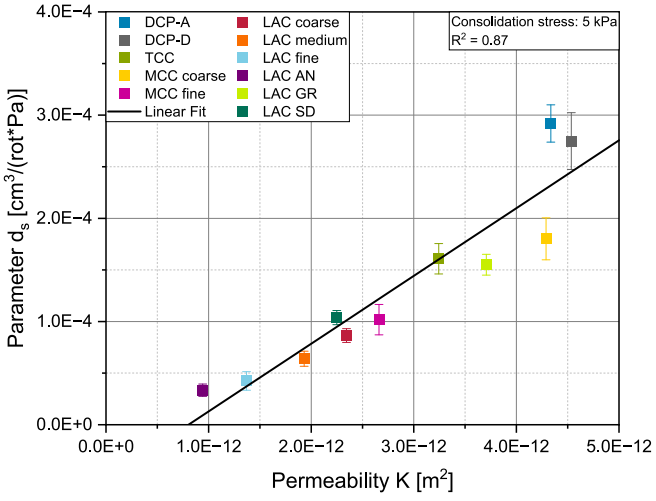


Fig. 11. Correlation between the parameter d_s and the permeability to describe the rotation-specific volume flow.

possible.

Following, $V_{r,s}$ can be expressed as followed:

$$V_{r,s} = (d_1 \cdot K - d_2) \cdot \Delta p \tag{22}$$

4.5. Predictive power of the model

The predictive quality of the developed model for suction filling is evaluated in the following. The model consists of two parts: first, the reference volume flow (Section 3.4.1) and second, the rotation-specific fill volume (Section 3.4.2), both depending on the differential pressure. The complete model results from combining eqs. (5), (15), (19) and (22):

$$\dot{V}_s = (d_1 \cdot K - d_2) \cdot \frac{p_0 \cdot v_p}{c + v_p} \cdot n_p + (a_1 \cdot K + a_2) \cdot \frac{p_0 \cdot v_p}{c + v_p} + b_1 \cdot \dot{V}_{Orifice} + b_2 \tag{23}$$

This equation enables the calculation of an estimated powder volume flow rate $\dot{V}_s$ into dies based on measurable powder properties and process parameters: permeability, powder flow through an orifice and the paddle speed. The model is developed based on data from eleven different materials with varying particle size, density, and flowability in order to allow a broad applicability to other powders. However, as the model is kept simple with multiple linear correlations, not every material might be represented best by the linear correlation with permeability and the volume flow through an orifice.

For a defined tableting problem that determines the necessary volume flow rate $\dot{V}_s$, the eq. (23) can be solved for different variables, for example, to calculate how well the powder needs to flow or how steep the fill cam has to be in order to provide a sufficient suction effect. In order to investigate the predictive quality of the developed model for the eleven materials on both rotary presses, the predicted critical paddle speed $n_{p,crit,pred}$ is compared with the experimentally obtained $n_{p,crit,\ exp}$. The critical paddle speed is the minimal required speed of the paddle, which ensures complete die filling at a given turret speed and can be calculated according to eq. (6). For industrial applications, calculating the optimal paddle speed is highly relevant. Wünsch et al. showed that excessively high speeds within the feed frame can cause intense mechanical stress on the powder bed, potentially affecting bulk properties like the flow behavior [26]. Similar findings were also stated by de Backere et al. [27]. Possible consequences also include particle segregation, mechanical breakage of sensitive particles or over-lubrication [28,29]. The predicted critical paddle speed is determined by the developed correlations for $\dot{V}_{0,s}$ (Eq. (19) and $V_{r,s}$ (Eq. (22)) with the respective powder properties permeability K and flow through an orifice $\dot{V}_{Orifice}$. For the experimentally determined values of $n_{p,crit,exp}$, $\dot{V}_{0,s}$ and $V_{r,s}$ are extracted from the initial fits of the volume flow rates. Graphically, this would be the paddle speed at the intersection between the linear fit of the powder flow rates into the dies and the corresponding minimum volume flow for complete die filling (comp. Fig. 4). Predicted and experimentally determined values are compared pairwise in parity plots for X 3 and XL 400 for both directions of rotation of the paddle wheel (Fig. 12).

Some materials fill the dies completely even at low turret speeds, resulting in only a very slight dependency between an increase in paddle speed and a further increase in the volume flow rate. This can be observed, for both the X3 and XL 400 with LAC AN. It has the lowest permeability, which means that, according to eq. (15), the highest negative pressure builds up and the strongest suction effect occurs. Accordingly, the filling is only minimally dependent on the paddle wheel speed and the experimentally determined critical paddle speeds are relatively low. However, the modeled paddle speeds according to eq. (23) are negative, which indicates that the volume flow solely based on the differential pressure already exceeds the necessary dosing volume. This does not seem reasonable in the context and is a result of the cumulative errors along the model development. The model is also not applicable to the MCC fine at the X3, as densification inside the dies occurs at low turret speed. Due to the ability of the MCC to be densified relatively easily, this densification could have already taken place inside the feed frame before the dies were filled. The densification could have been caused by the intensive rotation of the paddle wheel and the stress on the powder between the spokes and the lid or base plate of the feed frame. The powder, which was partially pre-densified beyond its bulk density, could then have entered the die. Respective data sets were already excluded for model development (Table 4), as the goal was to develop the model based on incomplete die filling.

In general, the critical paddle speed can be predicted with a medium accuracy. Deviations of the modeled value from the experimental one are mostly below 20 %. For the XL 400, slight deviation in terms of too low predicted paddle speeds can be identified for DCP-A and DCP-D, while values for LAC fine ccw are slightly overestimated as also for

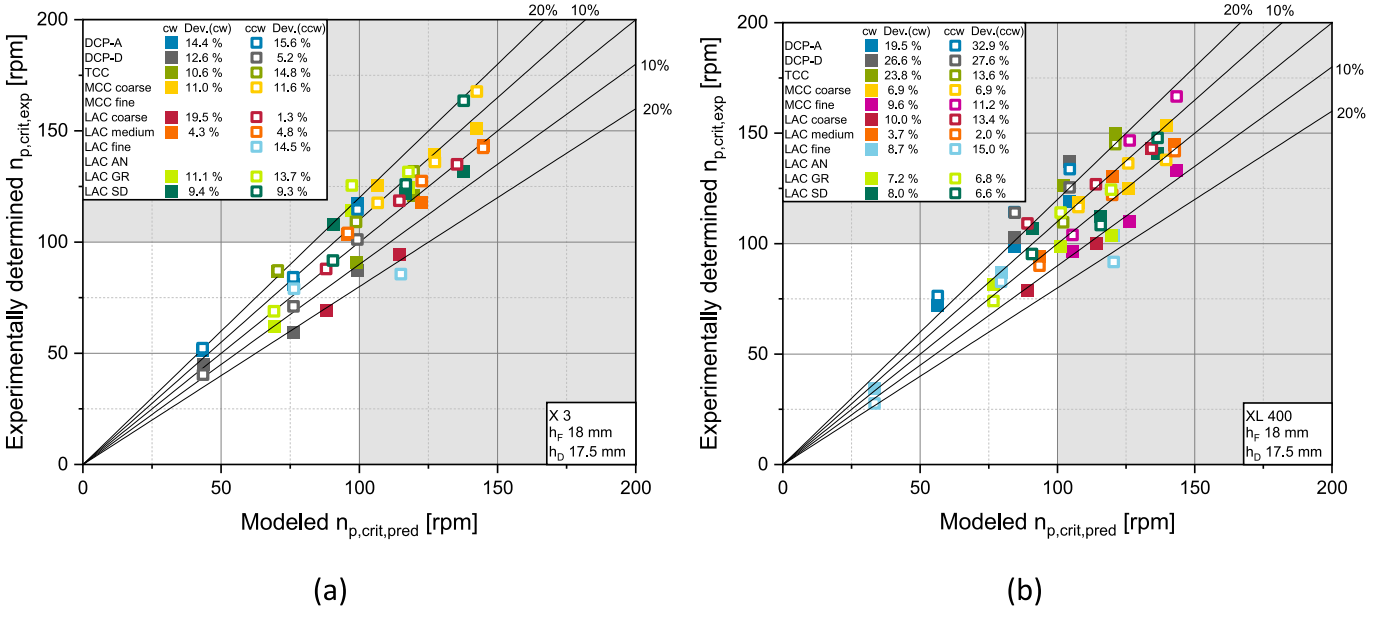


Fig. 12. Parity plots to show the deviations between the modeled critical paddle speed $n_{p,crit,pred}$ and the experimentally determined critical paddle speed $n_{p,crit,exp}$ for the X 3 (a) and the XL 400 (b); the relative deviations from the model were represented as mean values for each material and rotational direction.

the X 3. For the X 3, underestimated critical paddle speeds can be found for LAC GR cw, for example. Therefore, no systematic deviations for the rotary press or the direction of rotation can be identified.

For gravity filling, $n_{p,crit,pred}$ is more scattered for the direction of rotation than for suction filling, which highlights the strong impact of suction on filled powder weight despite other influencing factors, leading to more uniform results for cw and ccw [13].

Differences in predictability between the rotary presses are small. They can be caused by diverging constructions of the feed frame. The clearance between the paddle wheel and the lid of the feed frame is more than twice the value of the XL 400 for the X 3, which can have a considerable impact on the hold-up and the powder flow, especially above the paddles. The potentially higher powder bed on the X 3 might enable higher volume flow rates into the dies due to the higher differential pressure, while on the XL 400 the volume flow rate benefits more from an increase in paddle speed. In addition, differences in fill cams directly influence the differential pressures during die filling and thereby impact the extent of the suction filling effect, with no clear trend observable in the parity plots. Beyond this, the filling process itself is characterized by complex and often not fully predictable dynamics resulting from the interaction of different paddle speeds and geometries. Material-specific interactions further add to this complexity, so that observed behaviors cannot always be traced back to individual powder properties alone. In order to proof the hypotheses, further investigations are needed. Such may comprise systematic evaluations of the effects of paddle heights with in line measurements of the actual powder bed height inside the feed frame and the analysis of the flow pattern.

In general, the modeling process shown here presents the challenge of transferring many effects into a framework of mathematical equations. In addition, the model is based on mathematical approximations of experimental data regarding the die filling and powder characterization, which bring along experimental uncertainties. One possible limitation in transferability in this model approach could be that only single components were examined and no mixtures of different excipients or with active ingredients. The mixing with magnesium stearate prior to the tableting process is not included in these considerations. Theoretically, the model should also be applicable to mixtures, as the mixtures also have a distinct flowability and permeability. In order to enable broad applicability, the materials examined were used to cover a wide range of powder properties. One potential challenge in applying the model could be its transferability to mixtures with widely varying particle sizes or densities. In such systems, depending on the individual contents, different packing densities may occur [30]. This is particularly relevant as the particle movement inside the feed frame may also have an influence on further segregation processes. The validation of the developed model with complex mixtures should therefore also be investigated in the future. Another major challenge lies in selecting an appropriate powder characterization method. None of the examined techniques fully replicates the conditions within the feed frame, where a pre-stressed, actively moved powder flows through a die, although powder flow through an orifice provides the closest similarity. The distinct dynamics occurring inside the feed frame increase the complexity of choosing a suitable characterization method. Furthermore, the paddle rotation exerts additional mechanical stress on the powder, leading to changes in bulk properties such as particle breakage, segregation or over-lubrication caused by the incorporated magnesium stearate. Therefore, direct comparison with the initially characterized material is only partially applicable and must be carefully considered when evaluating the accuracy of the developed model [28,29]. One possible source of error that was not taken into account during model development is the superimposition or sequential occurrence of suction filling and gravity filling. This might be particularly interesting for powders with higher permeability like DCP-A or DCP-D, which may achieve lower volume flows in suction filling and benefit from subsequent gravity filling. A simplified test rig has already shown that a phase of gravity filling may follow a phase of suction filling, when specific process parameter combinations are applied [7]. For this purpose, measurements of differential pressures were carried out directly in the die, which is not technically feasible in rotary presses at present state. This means that it is not possible to distinguish between gravity and suction filling and clearly attribute certain fractions of the overall volume flows to a specific filling mechanism.

All these deviations sum up throughout the whole model development and are reflected in the parity plots, making the precise prediction of the filling behavior of all eleven materials on both presses challenging.

Although the model has certain limitations and cannot fully capture the complexity of the die filling process, it is important to also consider its advantages. Its main strength lies in the use of only standard powder characterization methods and basic process parameters, while still delivering predictions with acceptable accuracy. This balance between simplicity and predictive ability makes the model particularly useful for formulation and process development, as it helps reduce the number of experimental trials required and thereby saves both time and material.

5. Conclusion

In this study, a model for powder volume flow rates into the dies was developed for suction filling. Eleven different excipients were systematically studied on two production-scale rotary presses. The parameters of this model were parameterized by independently analyzed powder properties to predict critical process parameters such as paddle speed.

The effect of the suction filling is based on the developing differential pressure, which directly correlates with the pull-down movement of the lower punch. Therefore, die filling on a rotary press can be crucially improved by higher differential pressures, which result from a steeper fill cam design.

Furthermore, the effect of differential pressure on filling performance depends on the powder's permeability. Due to the higher gas-particle interaction in powders with low permeability, only small air flows through the powder bed are achieved under given conditions, leading to an enhanced suction filling effect. However, formulations with fine particles showed lower permeabilities and could considerably better be filled under suction. This should especially be considered during multi-layer tableting, since only the first layer can be filled by suction filling. The coarser, more easily flowing formulation should be on the second layer, which can only be filled by gravity filling. A correlation of the initial volume flow ($\Delta p = 0$ Pa) with the powder flow through an orifice was found to consider such differences in flowability and provides an easily accessible and reliable measure for die filling predictions.

The predictive quality of the model showed a high accuracy for both rotary presses, X 3 and XL 400, and both directions of paddle rotation. The average error for most materials and process parameter settings is less than 20 %. As eleven materials with strongly varying properties were considered in this study, it can be emphasized, that the filling behavior of probably all powders can be sufficiently reflected by the developed model. For process development and optimization, this model can be helpful to predict the minimal paddle speed depending on the desired turret speed and thus can accelerate the development of tableting processes. For manufacturers, this means time and cost savings and therefore a more efficient tablet production. Following, fine-tuning of the paddle speed enables production with low weight variability.

In the future, systematic research is necessary to explore the effects of paddle wheel as well as feed frame design characteristics. Understanding these geometrical aspects will be crucial for improving the model's predictive accuracy and ensuring it can effectively handle a wide range of equipment configurations and process conditions. Additionally, modeling die filling should be expanded to feed frames with higher numbers of paddles, which are commonly used for single-layer tableting. Another aspect that will be the focus of further investigations is the modeling of mass variations. In addition to the

absolute filled masses, the mass variations are a particularly important measure of how stable a process is and are therefore highly relevant regarding the industrial application. This should also be described in terms of possible correlations with powder properties, process parameters and component geometries.

CRediT authorship contribution statement

Ann Kathrin Schomberg: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lars Wagner:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Jan Henrik Finke:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization. **Arno Kwade:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: (Arno Kwade reports financial support was provided by German Research Foundation (413141366). Jan Henrik Finke reports financial support was provided by German Research Foundation (525028949). Ann Kathrin Schomberg and Lars Wagner reports equipment, drugs, or supplies was provided by KORSCH AG. Ann Kathrin Schomberg and Lars Wagner reports equipment, drugs, or supplies was provided by Chemische Fabrik Budenheim KG. Ann Kathrin Schomberg and Lars Wagner reports equipment, drugs, or supplies was provided by DFE Pharma. Ann Kathrin Schomberg and Lars Wagner reports equipment, drugs, or supplies was provided by J Rettenmaier and Sons. Ann Kathrin Schomberg and Lars Wagner reports equipment, drugs, or supplies was provided by Jungbunzlauer Ladenburg GmbH. Ann Kathrin Schomberg and Lars Wagner reports equipment, drugs, or supplies was provided by Peter Greven GmbH & Co KG. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.powtec.2025.121629.

Data availability

Data will be made available on request.

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Corrigendum to "Development of a model to predict die filling under suction on different rotary tablet presses" [Powder Technology Volume 468 (2026) 121629]

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The authors regret the following mistakes:

There are two transposed digits. The first one is on page 9 (after eq. 18), where the parameter b_2 is 0.111 instead of 0.011. The second one is on page 10 (after eq. 21), where the parameter d_2 is $5.271*10^{-5}$ cm^3/(rot*Pa) instead of $5.271*10^{7}$ cm^3/(rot*Pa). The calculations and further results and discussions are based on the correct values; they were just stated incorrectly.

The authors would like to apologise for any inconvenience caused.

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